

Research Memo - Dominant Transportation Imbalance

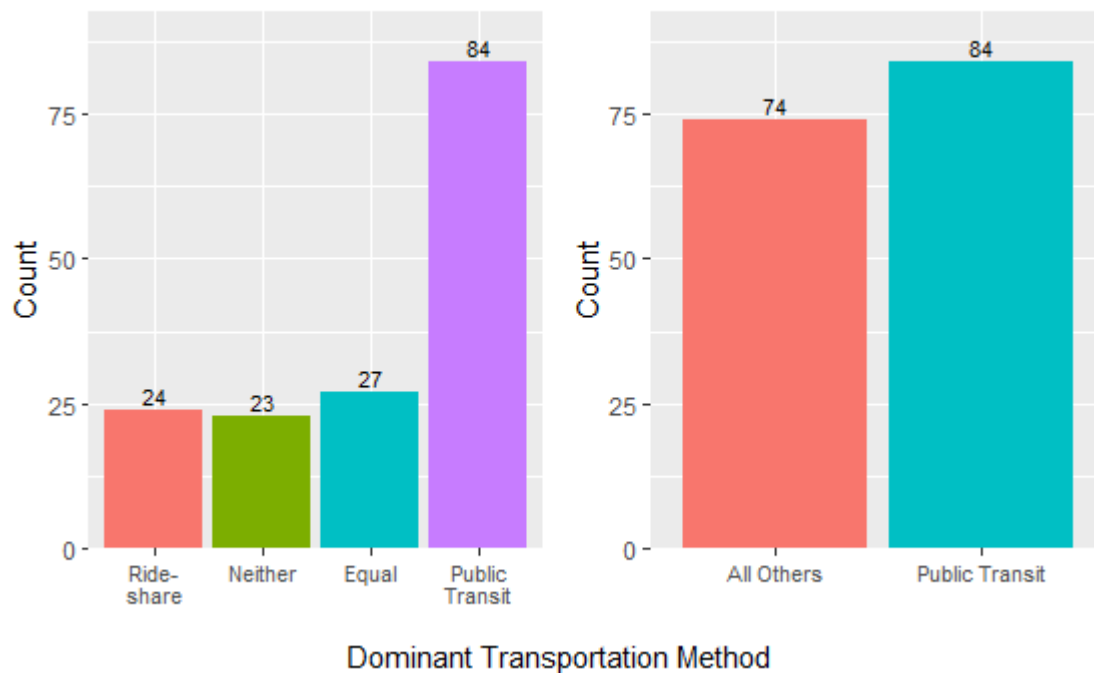
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Re: [Is the distribution of Dominant Transportation Methods balanced?]

The distribution of Dominant Transportation Methods is grossly imbalanced when considering all possible categories (rideshare, public transit, neither, equal), however the distribution becomes roughly balanced when collapsing categories down to public transit vs. all other possibilities combined.

Dominant Transportation Method is Imbalanced



These figures illustrate the dominance of public transit use is far more prevalent than all other categories, even when combined. The right-hand plot is a direct comparison of public transit dominance versus the combined count of rideshare, neither, and equal use respondents.

Attempting to build a 4-way classifier with the current dataset will prove difficult due to the limited sample size for our minority categories, and the challenges of resampling procedures with the inherently complicated natural language data we are using as inputs. When categories are collapsed as seen in the right-hand plot, however, our data immediately becomes much more balanced and thereby preferable for training a classifier. For these reasons, my recommendation is to pursue a **binary classification** of whether or not a respondent primarily uses public transit.

NOTE: The engineering of 'Dominant Transportation Method' occurs in the `clean_data.qmd` script.

NOTE: Rendering of the above figures occurs in the `eda_dependents.qmd` script.